

pgbadger Runbook

Here the link to the video of the runbook simulation.

Intro

pgbadger is a free tool that can help to generate a fast interactive HTML report from PostgreSQL's logs, it is very useful when you want to analyze logs with a lot of info. The report from pgbadger is about queries (it is always logged when the query time is greater than `log_min_duration_statement` parameter), connections, vacuum, lock, errors and others stuff

Use in GitLab

The first step is to detect the values of two PostgreSQL's parameters, in case of GitLab are the following:

```
show log_destination;
    log_destination
-----
    csvlog
(1 row)

show log_line_prefix;
          log_line_prefix
-----
 %m [%p, %x]: [%l-1] user=%u,db=%d,app=%a,client=%h
```

The following flags of pgbadger to analyze logs are depending on the previous result:

```
--format csv
--prefix '%m [%p, %x]: [%l-1] user=%u,db=%d,app=%a,client=%h'
--csv-separator ','
```

Example of use: The GitLab postgresQL's log are located in `/var/log/gitlab/postgresql`

```
#copy log files from DB server
```

```
scp -F ~/.ssh/ssh-file patroni-N-db-gprd.c.gitlab-production.internal:/var/log/gitlab/postgresql
```

```
###If not have permission with your user, can copy to your home and then copy scp command on
```

```
# zip files (zcat)
```

```
ssh -F ~/.ssh/ssh-file patroni-N-db-gprd.c.gitlab-production.internal "sudo zcat /var/log/gi
```

```
# plain files (cat)
```

```
ssh -F ~/.ssh/ssh-file patroni-N-db-gprd.c.gitlab-production.internal "sudo cat /var/log/gi
```

```
# analyzing 1 log file, but you can analyze more passing the names at the end
```

```
pgbadger --format csv --prefix '%m [%p, %x]: [%l-1] user=%u,db=%d,app=%a,client=%h' -o log_r
```

most_freq_queries

Figure 1: most_freq_queries

detail_query

Figure 2: detail_query

The output report will be in html file named `log_report.html`

The highlighted sections of the report

Queries reports:

Top

The top tab, is one of the most important tabs, and show the information about slowest queries found in the logs, see the sections called:

- Time consuming queries
- Most frequent queries

If make Click the button **Details** will see more info about this query related with count and duration

Queries

The Queries tab show the statistical information about queries found in the logs, see the sections called:

- Queries by type
- Queries by host

General reports

Overview

The overview tab shows some general statistical information, see the sections called:

- Global stats
- General Activity

query_by_type

Figure 3: query_by_type

global_stats

Figure 4: global_stats

general_activity

Figure 5: general_activity

Connections

The connections tab shows the graphs about connection information found in the logs, see the sections called:

- Connections per user
- Connections per host

Write reports:

Checkpoints

The checkpoints tab show the statistical information about the checkpoint process found in the logs, see the section called: **Checkpoints Activity**

Vacuums

The vacuums tab shows the information about VACUUMs and ANALYZEs occurring in the database found in the logs, see the sections called:

- Analyzes per table
- Vacuums per table
- Autovacuum Activity

Locks and Events reports:

locks

The locks tab show the information about locks occurring in the database found in the logs, see all section here lock_type

Events

The events tab show the anything information no show in another tabs, see the section called:

- Most Frequent Errors/Events

conn_per_host

Figure 6: conn_per_host

checkpoint_Activity

Figure 7: checkpoint_Activity

vacuum_per_table

Figure 8: vacuum_per_table

most_freq_error

Figure 9: most_freq_error